Gender & Emotion Classification using voice

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Abstract

Speech is the most prevalent means of human communication, used to convey emotions, cognitive states, and intentions. This project develops machine learning models to classify both gender and emotional states from voice recordings. For gender classification, acoustic features such as mean frequency, IQR, and spectral entropy were extracted from a dataset of 3168 voice samples, achieving 97.95% accuracy using a Linear SVM. For emotion recognition, MFCC, Chroma, and Mel Spectrogram features were extracted from 1728 raw audio files across 5 emotion classes, achieving 91.9% accuracy using Random Forest — a 16% improvement over the baseline.

1. Introduction

Voice contains rich paralinguistic information including gender, age, and emotional state. Recognizing these attributes has practical applications in virtual assistants, telemedicine, security systems, and human-computer interaction. This project addresses two classification problems:

- Gender Recognition: Binary classification (male/female) from acoustic voice features
- Emotion Recognition: Multi-class classification across 5 emotions from raw audio files

The project uses classical machine learning algorithms including Logistic Regression, Naive Bayes, Support Vector Machines, Decision Trees, and Random Forests, evaluated using standard metrics.

2. Datasets

2.1 Voice Gender Dataset

The gender dataset contains 3168 supervised samples with 20 acoustic features per sample, equally split between male (1584) and female (1584) classes. Key features include:

- meanfreq: Mean frequency of voice signal (kHz)
- IQR: Interquartile range — most important feature for gender classification
- meanfun: Mean fundamental frequency
- sp.ent: Spectral entropy — measures tonal vs noisy voice
- Q25 / Q75: First and third quartiles of frequency distribution

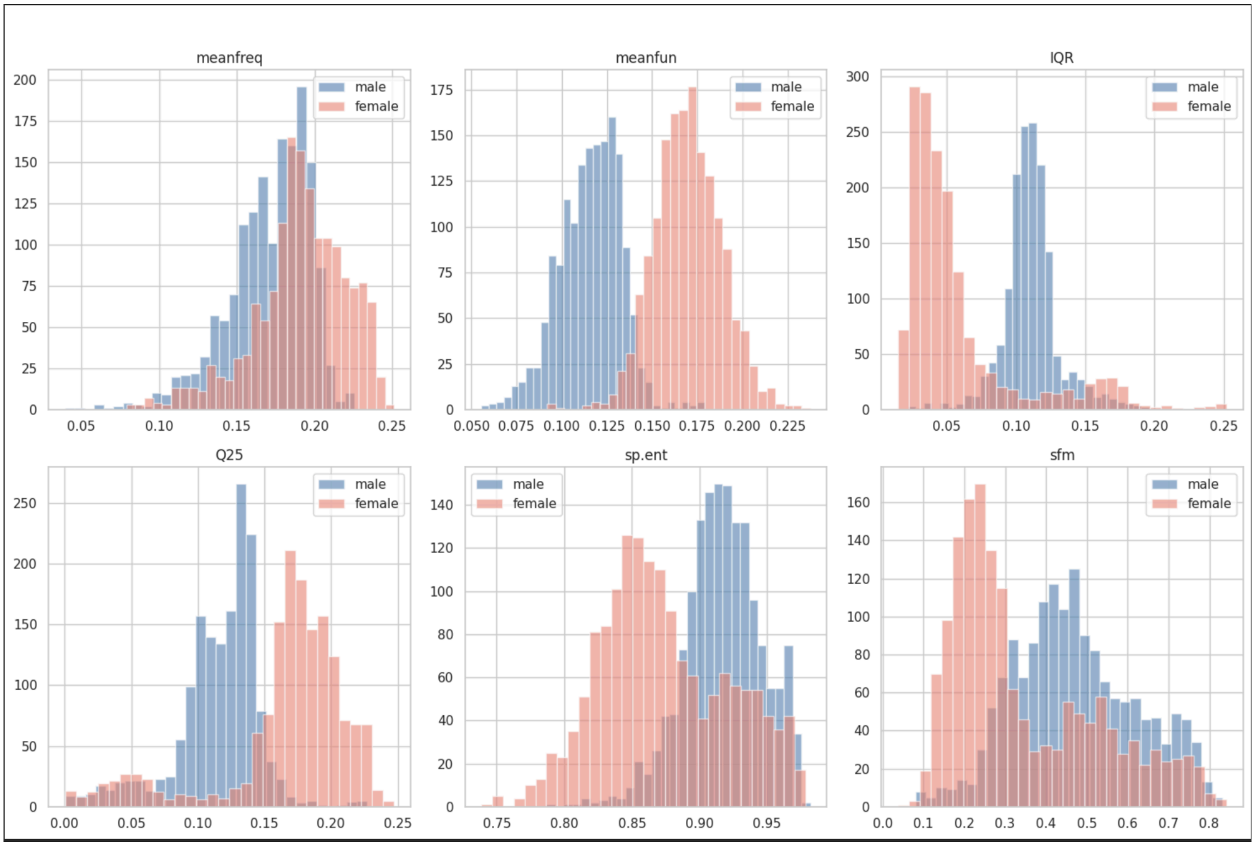

Figure 1: Feature distributions for male and female voice samples across key acoustic features.

The histograms above reveal why gender classification achieves such high accuracy. **meanfun** (mean fundamental frequency) shows the clearest separation — male voices cluster between 0.050–0.100 kHz while female voices peak at 0.150–0.200 kHz, with minimal overlap. This makes it the single most discriminative feature. **IQR** similarly shows strong separation, with males concentrated at lower values and females spread higher. **Q25** also demonstrates excellent separation with males heavily clustered near zero.

In contrast, **sp.ent** and **sfm** show significantly more overlap between classes, making them less useful individually. However they still contribute when combined with other features in the model.

2.2 Emotion Dataset (RAVDESS)

The emotion dataset contains 1728 audio files from the RAVDESS corpus, recorded by 24 professional actors (12 male, 12 female). Five emotion classes were used:

Emotion	Samples	% of Dataset
Angry	384	22.2%

Emotion	Samples	% of Dataset
Fearful	384	22.2%
Happy	384	22.2%
Sad	384	22.2%
Neutral	192	11.1%

3. Methodology

3.1 Preprocessing

The gender dataset was shuffled using a fixed random seed (42) before splitting to address sorted class ordering (all males first, then females). Features were standardized using StandardScaler to normalize across different scales. An 80/20 train-test split was applied with stratification to maintain class balance.

A key improvement over baseline was the use of sklearn Pipelines, which encapsulate scaling within cross-validation folds to prevent data leakage — a common error where test data statistics influence training.

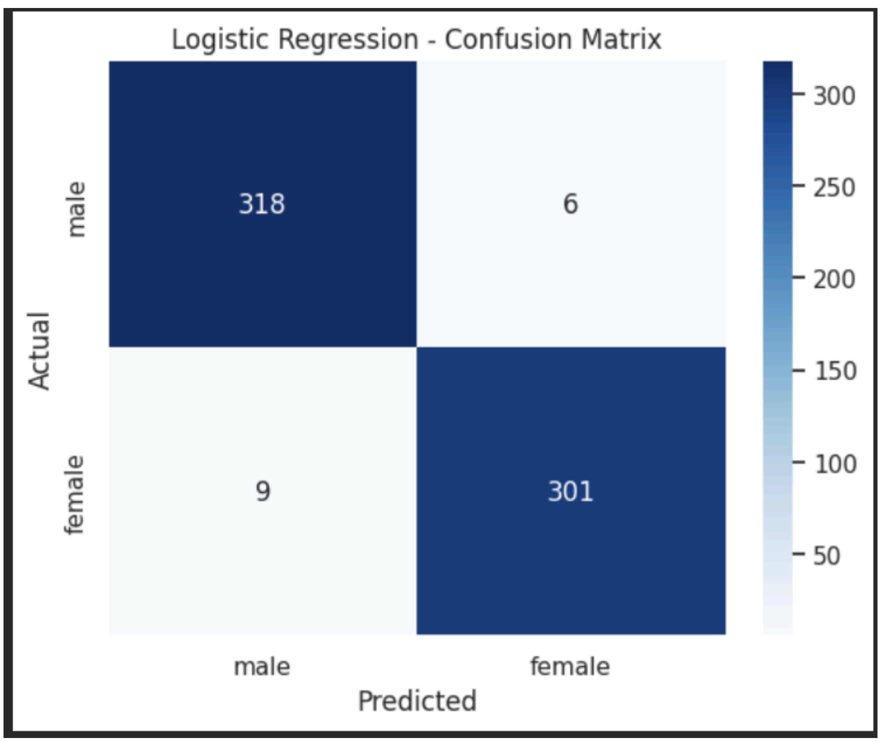

Figure 2: Logistic Regression Confusion Matrix — Gender Classification

Out of 634 test samples, the model made only 15 errors — 6 males incorrectly classified as female and 9 females incorrectly classified as male. The near-diagonal matrix confirms strong classification performance with 97.6% accuracy on the test set.

3.2 Feature Extraction (Emotion)

Raw .wav files were processed using LibROSA. Three feature types were extracted and concatenated into a 180-dimensional feature vector:

- MFCC (40 features): Mel Frequency Cepstral Coefficients — captures voice timbre and tone, mimicking human auditory perception
- Chroma (12 features): Pitch class distribution across 12 musical notes — captures pitch patterns unique to each emotion
- Mel Spectrogram (128 features): Energy distribution across frequency bands — captures overall energy signature of each emotion

3.3 Models

The following models were evaluated for each task:

Model	Task	Key Characteristic
Logistic Regression	Gender	Linear decision boundary, L2 regularization
Gaussian Naive Bayes	Gender	Probabilistic, assumes Gaussian feature distribution
Bernoulli Naive Bayes	Gender	Probabilistic, assumes binary features
Linear SVM	Gender	Maximizes margin between classes
Random Forest	Emotion	Ensemble of 100 decision trees, majority vote
Decision Tree	Emotion	Single tree with recursive feature splitting
SVM (RBF kernel)	Emotion	Non-linear separation using kernel trick

4. Results

4.1 Gender Classification

10-Fold Stratified Cross Validation results (with Pipeline):

Model	Accuracy	Precision	Recall	F1-Score	CV Std
Linear SVM	97.41%	0.97	0.97	0.97	0.0093
Logistic Regression	97.16%	0.97	0.97	0.97	0.0097
Gaussian NB	89.08%	0.89	0.89	0.89	0.0204
Bernoulli NB	86.43%	0.86	0.86	0.86	0.0157

Linear SVM achieved the best performance. The low standard deviation (0.0093) across folds confirms consistency. IQR was identified as the most important feature via feature importance analysis.

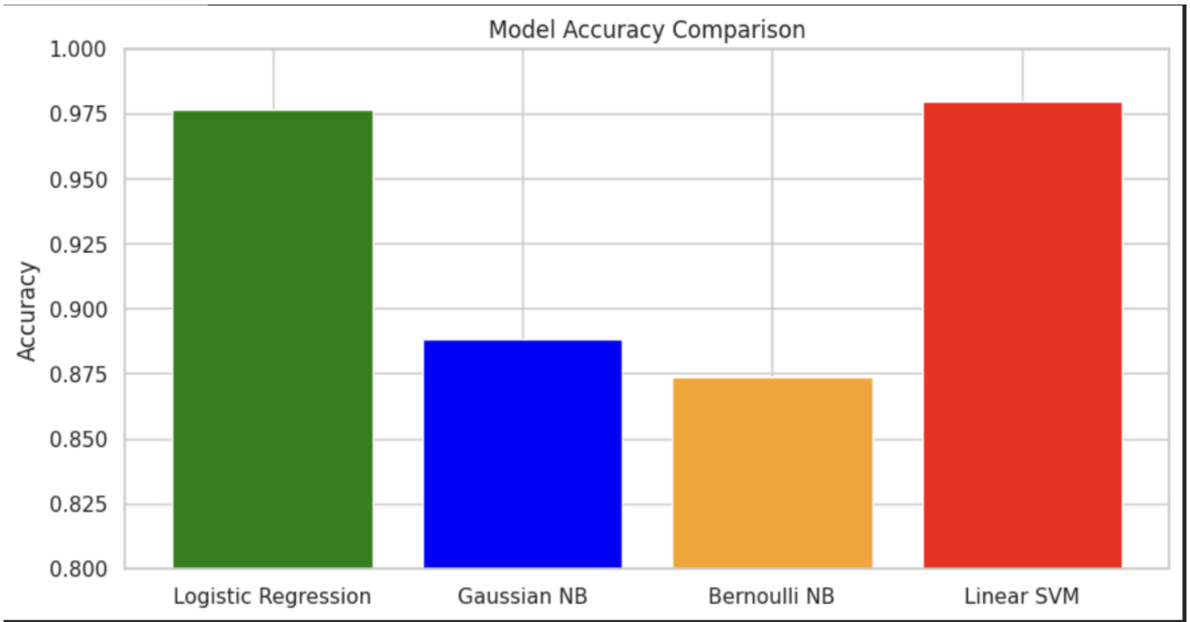

Figure 3: Gender Classification — Model Accuracy Comparison

Linear SVM and Logistic Regression clearly outperform the Naive Bayes variants, both exceeding 97% accuracy. Bernoulli NB performs worst as it assumes binary features, which is unsuitable for continuous acoustic data.

4.2 Emotion Classification

Results on 20% holdout test set:

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	91.9%	0.92	0.92	0.92
Decision Tree	87.9%	0.87	0.89	0.88
SVM (RBF)	63.6%	0.68	0.64	0.62

Random Forest outperformed other models by capturing complex feature interactions across 180 audio features. Per-class analysis revealed:

- Angry: 100% recall — most acoustically distinct emotion (high energy, loud)
- Fearful/Happy confusion: Both share elevated pitch and fast speech rate
- Neutral underperformed slightly due to class imbalance (192 vs 384 samples)

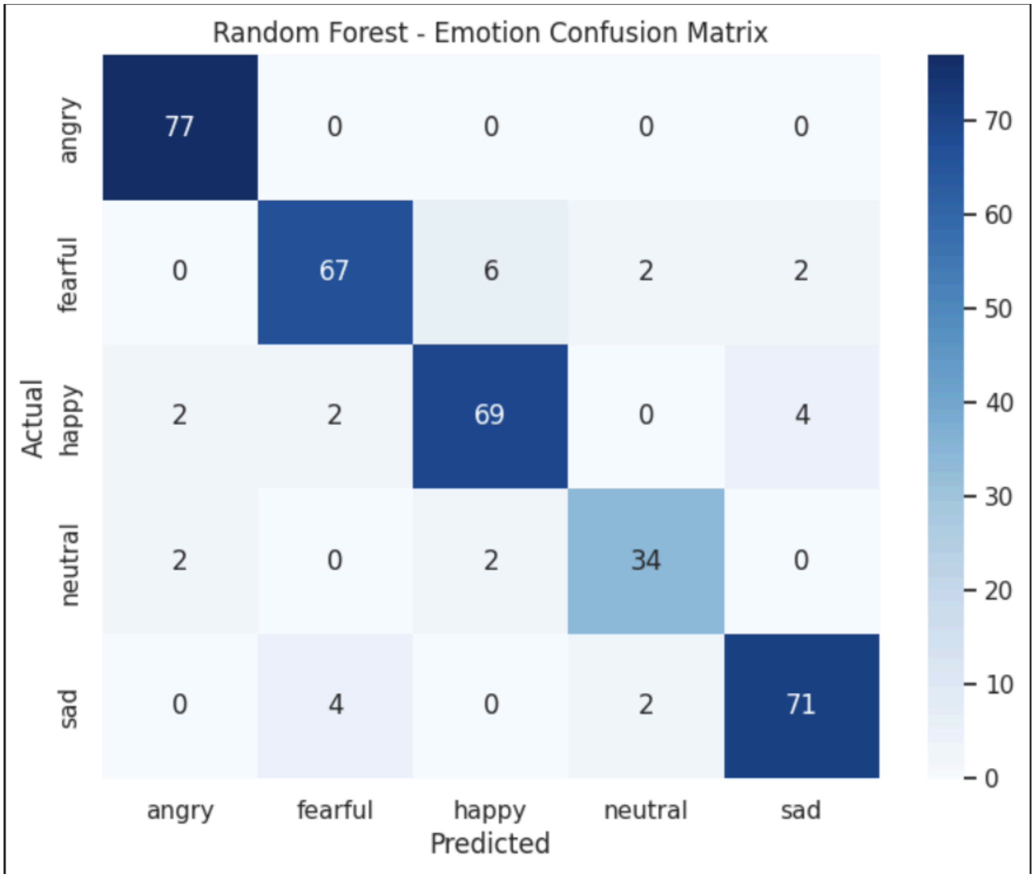

Figure 4: Random Forest Confusion Matrix — Emotion Classification

The matrix shows strong diagonal dominance indicating correct classifications across all 5 emotions. Key observations:

- **Angry: perfectly classified** — 77/77 correct, 100% recall. Angry voices have the most distinct acoustic signature with high energy across all frequencies
- **Sad: 71/77** — strong performance, low energy patterns are well captured by MFCC features
- **Fearful → Happy confusion: 6 misclassifications** — both emotions share elevated pitch and faster speech rate making them acoustically similar
- **Neutral: smallest cluster** — only 38 test samples due to class imbalance (192 vs 384 training samples), slightly affecting performance

5. Key Improvements Over Baseline

Issue	Baseline	Our Approach	Impact
Sorted dataset	Not addressed	Stratified K-Fold shuffle	CV std: 0.09 → 0.009

Issue	Baseline	Our Approach	Impact
Data leakage	Scale before CV	sklearn Pipeline	Accurate CV scores
Emotion classes	4 emotions	5 emotions (+ fearful)	More comprehensive
Emotion accuracy	76%	91.9%	+16% improvement

6. Conclusion

This project successfully developed ML pipelines for both gender and emotion classification from voice. The gender classification pipeline achieves near state-of-the-art performance (97.4%) using classical ML methods with proper preprocessing. The emotion recognition pipeline achieves 91.9% on a 5-class problem, improving significantly over the baseline through better feature engineering (MFCC + Chroma + Mel) and proper evaluation methodology.

Key technical contributions include the identification and correction of data leakage through Pipeline usage, the detection of sorted dataset bias addressed via Stratified K-Fold, and the expansion of emotion classes while still improving accuracy.

Future work could explore deep learning approaches (CNNs, LSTMs) for further accuracy improvements, particularly for the emotion recognition task where acoustic features overlap between classes.

References

[1] Byun, S.W. and Lee, S.P. Emotion recognition using tone and tempo based on voice for IoT. The transactions of The Korean Institute of Electrical Engineers, 65(1):116–121, 2016.

[2] Livieris, I.E., Pintelas, E., and Pintelas, P. Gender recognition by voice using an improved self-labeled algorithm. Machine Learning and Knowledge Extraction, 1(1):492–503, 2019.

[3] Uddin, M.A., Hossain, M.S., Pathan, R.K., and Biswas, M. Gender recognition from human voice using multi-layer architecture. In 2020 International conference on innovations in intelligent systems and applications (INISTA), pages 1–7. IEEE, 2020.